a. Working principle

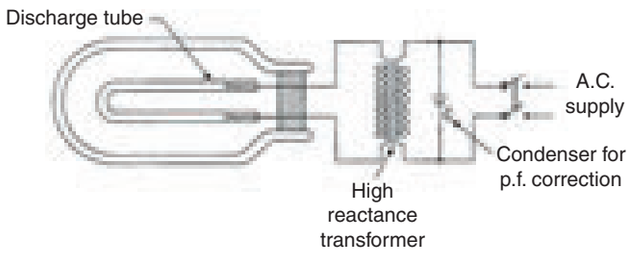


Fig. 2.6 Sodium vapour lamp

The construction of sodium vapour lamp is shown in figure 2.6. Electric discharge lamps require a high voltage at start and a low voltage during operation. The tungsten-coated electrodes are connected across auto-transformer, having high leakage reactance. The open-circuit voltage of this transformer is about 450 V which is sufficient to initiate a discharge through the neon gas.

After 10 to 15 minutes, the voltage falls to 150 V, due to low power factor. A capacitor is connected across the supply to improve the power factor. The colour of light produced is yellowish.

b. Applications

1. LPS lamps are rarely used for indoor lighting and are best suited for outdoor lighting.
2. LPS lamps are used in security lighting as their high efficiency.
3. LPS lamps are also often used in long tunnels.

2.6.2 Mercury vapour lamp

On the basis of pressure inside the discharge tube, the mercury vapour lamps are classified as High pressure mercury vapour lamp and Low pressure mercury vapour lamp. High pressure mercury vapour lamps are classified as:

1. MA type (Mercury vapour lamp with auxiliary electrode)

These are operated at 220 - 250 volt, AC supply and manufactured in 250 to 400 watts.

2. MAT type (Mercury vapour lamp with tungsten filament)

These are manufactured between 300 to 500 watts and works at 200 to 250V (Both AC and DC)

3. MB type (Mercury vapour lamp with auxiliary electrode and bayonet cap)

This type is operated at 200 – 250 volt, (AC and made in 80 watts and 125 watts)

a. Construction

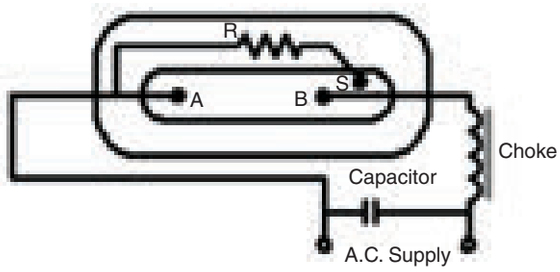


Fig. 2.7 Mercury vapour lamp

The construction of mercury vapour lamp is shown in figure 2.7. It consists of hard glass tube enclosed in outer bulb of ordinary glass. The space between two bulb is completely evacuated to prevent heat loss by convection from the inner bulb. The outer bulb absorbs harmful ultraviolet rays. The inner bulb contains argon gas with a certain quantity of mercury. In addition to two electrodes, starting electrode having high resistance, connected in series is also provided. The main electrodes are made of tungsten wire in a helical shape. The lamp has a screwed cap and is connected to supply with a choke. A capacitor is connected across supply to improve power factor.

BEE_ENG_Chapter_02.indd 26 12/27/2019 10:18:33 AM